

Let G be an abelian group. We define a character on G as a homomorphism

$$\phi: G \rightarrow \mathbb{C}^*$$

If G is a finite abelian group then $|\phi(g)| = 1 \forall g \in G$
 $|G| = n$

$$\text{Then } |g^n| = 1 \forall g \in G$$

$$\Rightarrow |\phi(g)|^n = 1$$

$$\Rightarrow |\phi(g)| = 1$$

Let V be the set of all complex valued fn's on a finite abelian group G .

$$\langle \phi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \phi(g) \overline{\psi(g)}$$

Let ϕ, ψ be two characters on G .

Then $\langle \phi, \psi \rangle = 0$ if $\phi \neq \psi$

Lemma: Let $\phi: G \rightarrow \mathbb{C}^*$ be a non-trivial character.

$$\text{Then } \sum_{g \in G} \phi(g) = 0$$

Proof: Let $S = \sum_{g \in G} \phi(g)$

Take $g_0 \in G$ such that $\phi(g_0) \neq 1$

$$\phi(g_0) S = \phi(g_0) \sum_{g \in G} \phi(g) = \sum_{g \in G} \phi(g_0 g) = S$$

$$\text{Since } \phi(g_0) \neq 1, S = 0$$

Proposition: Let G be a finite cyclic group.

Then the number of characters on the group G is

$|G|$.

Proof: Let $G = \langle g \rangle$. Then any group homomorphism $\phi: G \rightarrow \mathbb{C}^*$ is given by mapping g to an n th root of unity.

$\hat{G}$ denotes the group of all characters on a finite abelian group G .

$$|\hat{G}| = |G|$$

If G is a finite abelian group then $\hat{G}$ is the set.

$$\bullet |\hat{G}| = |G|$$

Fundamental theo. on fin. abelian group:-

$$G = \mathbb{Z}/n_1\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/n_k\mathbb{Z}$$

By univ. prop. $\circ$

$$\hat{G} \cong (\hat{\mathbb{Z}}/n_1\mathbb{Z}) \oplus \dots \oplus (\hat{\mathbb{Z}}/n_k\mathbb{Z})$$

Let G be finite abelian group. $|G| = n$

$e_1, \dots, e_n$ be n distinct chars on G .

Let f be a complex valued function on the group G .

Let $\hat{f}_k = \langle f, e_k \rangle$, Then $f = \sum_{k=1}^n \hat{f}_k e_k$

$\bullet$ Let $\{a_n\}_{n \in \mathbb{N}}$ be a sequence $a_n \in \mathbb{C}$. We say that $\prod_{n=1}^{\infty} a_n$

converges if $\lim_{n \rightarrow \infty} \prod_{k=1}^n a_k$ exists.

or equivalently, $\forall \varepsilon > 0, \exists m \in \mathbb{N}, n$ such that $a \in \mathbb{C}$

$$\forall n \geq m$$

$$\left| \prod_{k=1}^n a_k - a \right| < \varepsilon$$

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \quad , s > 1$$

$$\zeta(s) = \prod_{p \text{ prime}} \left(\frac{1}{1 - \frac{1}{p^s}} \right)$$

Later on, using the formula, Euler proved that

$$\sum_{p \text{ prime}} \frac{1}{p} \text{ diverges}$$

A Dirichlet character modulo q , is a

$$\phi: \mathbb{Z} \rightarrow \mathbb{C}^* \text{ such that}$$

$$(i) \quad \phi(ab) = \phi(a)\phi(b)$$

$$(ii) \quad \phi(a) = 0 \text{ if } \gcd(a, q) \neq 1$$

$$(iii) \quad \phi(a+q) = \phi(a)$$

A Dirichlet character is an extension of a character on the group $\mathbb{Z}^*_q$ to $\mathbb{Z}$

Let χ be a Dirichlet character.

$$L(s, \chi) = \sum_{n=1}^{\infty} \chi(n)/n^s$$

We need to define some logarithm functions on

$$\mathbb{C}, \quad \log \left(\frac{1}{1-z} \right) = \sum_{n=1}^{\infty} \frac{z^n}{n} \quad \text{where } |z| < 1$$

Proposition: Let $|z| < 1$

$$\text{Then } i) \quad \log \left(\frac{1}{1-z} \right) = \frac{1}{1-z}$$

$$ii) \quad \text{If } |z| < 1/2$$

$$\log z = z + E(z)$$

$$|E(z)| \leq |z|^2$$

ciii) If $|z| < 1/2$. Then $|\log(\frac{1}{1-z})| \leq 2|z|$

Pf: (ii) $\log(\frac{1}{1-z}) = \sum_{n=1}^{\infty} \frac{z^n}{n}$

$$\log(\frac{1}{1-z}) - z = \sum_{n=2}^{\infty} \frac{z^n}{n}$$

$$= \frac{z^2}{2} \left(1 + \frac{z}{3/2} + \frac{z^2}{4/2} + \dots \right) = R$$

$$|R| \leq \left| \frac{z^2}{2} \right| \left(\sum_{n=0}^{\infty} |z|^n \right) \quad \text{absolutely}$$

$$\leq \left| \frac{z^2}{2} \right| \left(1 + \frac{1}{2} + \frac{1}{4} + \dots \right)$$

$$\leq |z^2|$$

Proposition: Let a_n be a sequence such that

$$\sum_{n=1}^{\infty} |a_n| < \infty$$

Then $\prod_{n=1}^{\infty} \frac{1}{1-a_n}$ converges

Since $\sum_{n=1}^{\infty} |a_n| < \infty$, $|a_n| \leq 1/2 \forall n \geq n_0$

$$\sum_{n=1}^{\infty} \left| \log\left(\frac{1}{1-a_n}\right) \right| \leq \sum_{n=1}^{\infty} |2a_n|$$

$$a = \sum_{n=1}^{\infty} \log \left(\frac{1}{1-a_n} \right) \text{ converges}$$

$$\theta \sum_{n=1}^{\infty} \log \left(\frac{1}{1-a_n} \right) = \theta a$$

$$\therefore \prod \left(\frac{1}{1-a_n} \right) \text{ converges}$$

Let χ be a Dirichlet character, $s > 1$

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$$

$$L(s, \chi) = \prod_{p \text{ prime}} \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right)$$

$$\sum_{\text{primes}} \left| \frac{\chi(p)}{p^s} \right| \leq \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$$



Dirichlet series
conv.

Therefore
LHS conv.

$\therefore$ By prev. Lemma ~~the~~ $\prod_{p \text{ prime}} \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right)$ conv.

$$\lim_{n \rightarrow \infty} \prod_{\text{primes}} \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right) = \prod$$

$$\prod_n = \prod_{\substack{p \text{ prime} \\ p \leq n}} \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right)$$

Stochastic Process: $\chi, \chi, \dots$

$$\prod_{n,m} = \prod_{\substack{p \leq n \\ \text{prime}}} \left(1 + \frac{\chi(p)}{p^s} + \frac{\chi(p^2)}{p^{2s}} + \dots + \frac{\chi(p^m)}{p^{ms}} \right)$$

Let S_n denote the n th partial sum in Dirichlet L series.

Then we can choose n large enough such that

$$|L(s, \chi) - S_n| < \varepsilon$$

$$|\prod - \prod_n| < \varepsilon$$

We can choose m to be large enough such that

$$|\prod_n - \prod_{n,m}| < \varepsilon$$

$$|S_n - \prod_{n,m}| < \varepsilon$$

$$\begin{aligned} |L(s, \chi) - \prod| &\leq |L(s, \chi) - S_n| \\ &\quad + |S_n - \prod_{n,m}| \\ &\quad + |\prod_{n,m} - \prod_n| \\ &\quad + |\prod_n - \prod| \end{aligned}$$

$L(s, \chi)$ is given by $< 4\varepsilon$

$$\prod_{\text{primes}} \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right)$$

Proposition: Let χ be a non-principal char.
Then $L(s, \chi)$ is cont. diff. for $0 < s < \infty$.

(ii) $\exists$ constants c, c', α and B such that

$$|L(s, \chi) - 1| < \alpha e^{-cs} \quad \text{as } s \rightarrow \infty$$

$$|L'(s, \chi)| \leq |B e^{-c's}|$$

$$\log \chi(s) = \int_s^\infty \frac{L'(t, \chi)}{L(t, \chi)} dt \quad \text{as } s \rightarrow \infty$$

$$\log \chi(s) = \sum_{p \text{ prime}} \log \left(1 - \frac{\chi(p)}{p^s} \right)$$

$$e^{\log \chi(s)} = L(s, \chi)$$

Theorem: χ be a non-trivial.

Then $L(1, \chi)$ is bounded & non zero

Fix a non-trivial character χ

$\lim_{s \rightarrow 1^+} \log \chi(s)$ exist

$$L(s, \chi) = \prod_{\text{primes}} \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right)$$

$$\log \chi(s) = \sum_{\text{primes}} \log \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right)$$

$$\log \left(\frac{1}{1 - \frac{\chi(p)}{p^s}} \right) = \frac{\chi(p)}{p^s} + E(p), \quad E(p) \leq \chi(p)^2 / p^{2s}$$

$\in \mathbb{N}$ such that

Principal character mod d
 $\chi_0(n) = \begin{cases} 1, & \text{if } \gcd(n, d) = 1 \\ 0, & \text{otherwise} \end{cases}$

$$\log \chi(s) = \sum_{p \text{ primes}} \frac{\chi(p)}{p^s} + E$$

$$|E| \leq \sum_{\text{prime}} \left| \frac{\chi(p^2)}{p^{2s}} \right| = \sum_{\text{prime}} \frac{1}{p^2} < \infty$$

$$\log \chi(s) = \sum_{\text{primes}} \frac{\chi(p)}{p^s} + E \text{ such that } |E| < \infty$$

$$\lim_{s \rightarrow 1} \sum_{\substack{p \text{ prime} \\ d \mid (p-1)}} \frac{1}{p^s} = \infty$$

If χ is trivial character then

$$\lim_{s \rightarrow 1} \sum_{\substack{p \\ \text{primes}}} \frac{\chi(p)}{p^s} = \infty$$

Else it is bounded.

$$a, a+d, a+2d, \dots \equiv a \pmod{d}$$

$$\text{Then } \delta_a(n) = \begin{cases} 1 & \text{if } n \equiv a \pmod{d} \\ 0 & \text{otherwise} \end{cases}$$

$$\delta_a = \sum_{\substack{\chi \\ \text{character}}} \frac{\overline{\chi(a)}}{|\mathbb{Z}^*_d|} \chi$$

In particular, we can write down the series

$$\sum_{\text{primes}} \frac{\delta_n(p)}{p^s} = \sum_{\text{primes}} \sum_{\chi} \frac{\overline{\chi(a)}}{|\mathbb{Z}^*_d|} \frac{\chi(p)}{p^s}$$

$$= \sum_{\chi} \left(\sum_{\text{primes}} \frac{\overline{\chi(a)}}{|Z^*(q)|} \frac{\chi(p)}{p^s} \right)$$

As a result, we get that

$$\sum_{\text{primes}} \frac{S_a(p)}{p} = \sum_{\substack{p \text{ prime} \\ p \equiv a \pmod{q}}} 1/p = \infty$$